SYSTEMS BIOLOGY GIVES CLUES ABOUT THE NEUROLOGICAL IMPAIRMENT IN MPSS

GERDA CRISTAL VILLALBA SILVA, URSULA MATTE
BIOINFORMATICS CORE, HCPA - UFRGS

INTRODUCTION

Mucopolysaccharidoses (MPS) are a group of rare genetic diseases. There are 11 types of MPS, and 7 of them present neurological symptoms.

Cognitive impairment · Sleep disturbances · Anxiety · Behavioral problems · Seizures · Hydrocephalus / macrocephaly · dementia · epilepsy · Depression

Variable according to the patient

METHODS

RStudio · edgeR · limma · GEO Gene Expression Omnibus · STRING · Cytoscape: MCODE (Clusters a given network), cytoHubba, BiNGO, CluePedia (CluePedia: A ClueGO plugin I and in silico data)

DATASETS

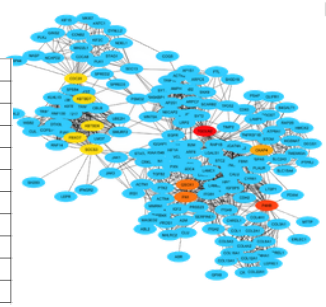
GSE CODE	MPS
GSE111906	MPS I
GSE95224	MPS II
GSE23075	MPS IIIB
GSE15758	MPS IIIB
GSE76283	MPS VII

For more information about the datasets, please visit our tool MPSBase:

RESULTS

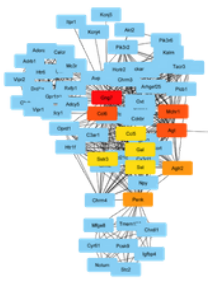
For instance, for MPS I, the related ontologies were clathrin derived vesicle budding and clathrin-mediated endocytosis. MPS II top genes were related to brain renin-angiotensin system. MPS IIIB top genes comprises the ontologies acetylation and chromatin regulation. The MPS IIIB top 10 genes were related to the activation of the mRNA cap-binding complex. Finally, MPS VII top genes were present in the ontologies C-MYB transcription factor network

A



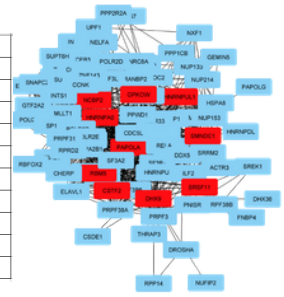
Rank	Node
1	TGOLN2
2	P4HB
3	FN1
3	QSOX1
5	CKAP4
6	CDC20
6	FBX07
6	KBTBD6
6	KBTBD7
6	SOCS3

B



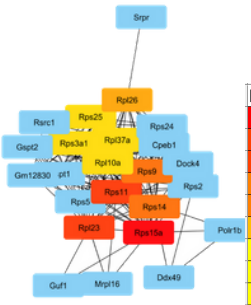
Rank	Node
1	GNG7
2	AGT
2	CD6
2	MCHR1
5	AGTR2
5	PENK
7	GAL
7	SSTR3
7	SSTR3
7	CD5

C



Rank	Node
1	GPKOW
1	RBM5
1	HNRNPUL1
1	SRSF11
1	DHX9
1	NCBP2
1	PAPOLA
1	CTSF2
1	HNRNPA0
1	SMNDC1

D



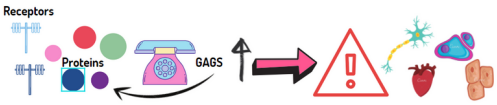
Rank	Node
1	RPS15a
2	RPS11
2	RPL23
4	RPS14
4	RPS9
6	RPL26
7	RPS25
7	RPL37a
7	RPS3a1
7	RPL10a

NETWORKS:
A = GSE111906, MPS I
B = GSE95224, MPS II
C = GSE23075, MPS IIIB
D = GSE15758, MPS IIIB
E = GSE76283, MPS VII

E

Rank	Node
1	SERPINA3N
2	LYZ1
2	LYZ2
4	TOLLIP
5	PYCARD
5	HMHA1
5	GNS
5	CYB5R3
5	GALNS
5	CTSC

Glycosaminoglycans interact with a wide range of proteins and receptors, several pathways are deranged in MPS!



CONCLUDING REMARKS

Systems biology approaches may help us to understand the mechanisms of neuropathology in the different types of Mucopolysaccharidoses, and to discover novel biomarkers and treatments for the neurological symptoms of these diseases.

PPGBM Programa de Pós-Graduação em Genética e Biologia Molecular · Hospital de Clínicas Porto Alegre RS · iSCB Poster Palooza · CNPq Conselho Nacional de Desenvolvimento Científico e Tecnológico